

Eversoft HRMS Integration Stack

A decoupled, asynchronous architecture designed to handle high-velocity event ingestion alongside heavy compute AI tasks.

Building for Scale

Mapping out the critical components for Webhooks, Queueing, AI
Compute, and Storage.

Webhook Ingestion & Gateway Layer

- **Web Framework: FastAPI (Python)**

Its native asynchronous design is perfect for absorbing high volumes of incoming webhooks without blocking.

- **Webhook Infrastructure: Hookdeck / Svix**

Acts as an Event Gateway to ingest LinkedIn and Naukri webhooks, handle backpressure, and route safely. Svix ensures secure, compliant outbound delivery.

- **Local Development: Hookdeck CLI / Ngrok**

Enables seamless routing of live Naukri and LinkedIn webhooks directly to your local machine for rapid testing.



Queueing & Async Processing Layer

Message Broker

Redis or RabbitMQ

Crucial for buffering incoming candidate data during massive traffic spikes. This ensures your system can gracefully absorb sudden loads so you never drop payloads from job boards.

Task Orchestration

Temporal or Celery

Temporal is particularly strong here because it handles stateful workflows—perfect for multi-step tasks like: Fetch Application → Download Resume → Extract Text → Run AI Score.

Resume Extraction (Pre-AI)

Before an AI can evaluate a candidate, you must extract clean text from messy PDFs or Word documents.

- **Document Parsing: Unstructured.io / LlamaParse**

These tools excel at extracting structured data from complex documents, handling multi-column resume layouts and tables effectively.

- **Enterprise Parsers: Sovren / MokaHR**

Provides a managed service with built-in recruitment ontologies for highly accurate, granular parsing APIs designed for HR tech.



The AI Match & Scoring Engine



Pipeline Orchestration

LangChain: Ideal for building the Retrieval-Augmented Generation (RAG) pipelines that retrieve the Job Description and compare it against the parsed resume text.



Inference Engine

vLLM (Docker/Linux): Process high volumes of resumes accurately without extreme cloud costs. A local setup on AMD hardware scales across nodes and GPUs.



Model Selection

Qwen2.5: Using a highly capable open-weight model allows complex reasoning for ranking without sending



System Integration

MCP Servers: Use Model Context Protocol to allow your AI engine to securely query your PostgreSQL database or

Data Normalization & Storage

> **Data Validation: Pydantic**

Use this to strictly enforce your candidate schema, coercing the wildly different JSON shapes coming from Naukri and LinkedIn into a single, unified Eversoft data model.

> **Database: PostgreSQL with pgvector**

A robust relational database for tracking the candidate pipeline, with the added benefit of vector storage for semantic similarity searches across candidate skills and job requirements.

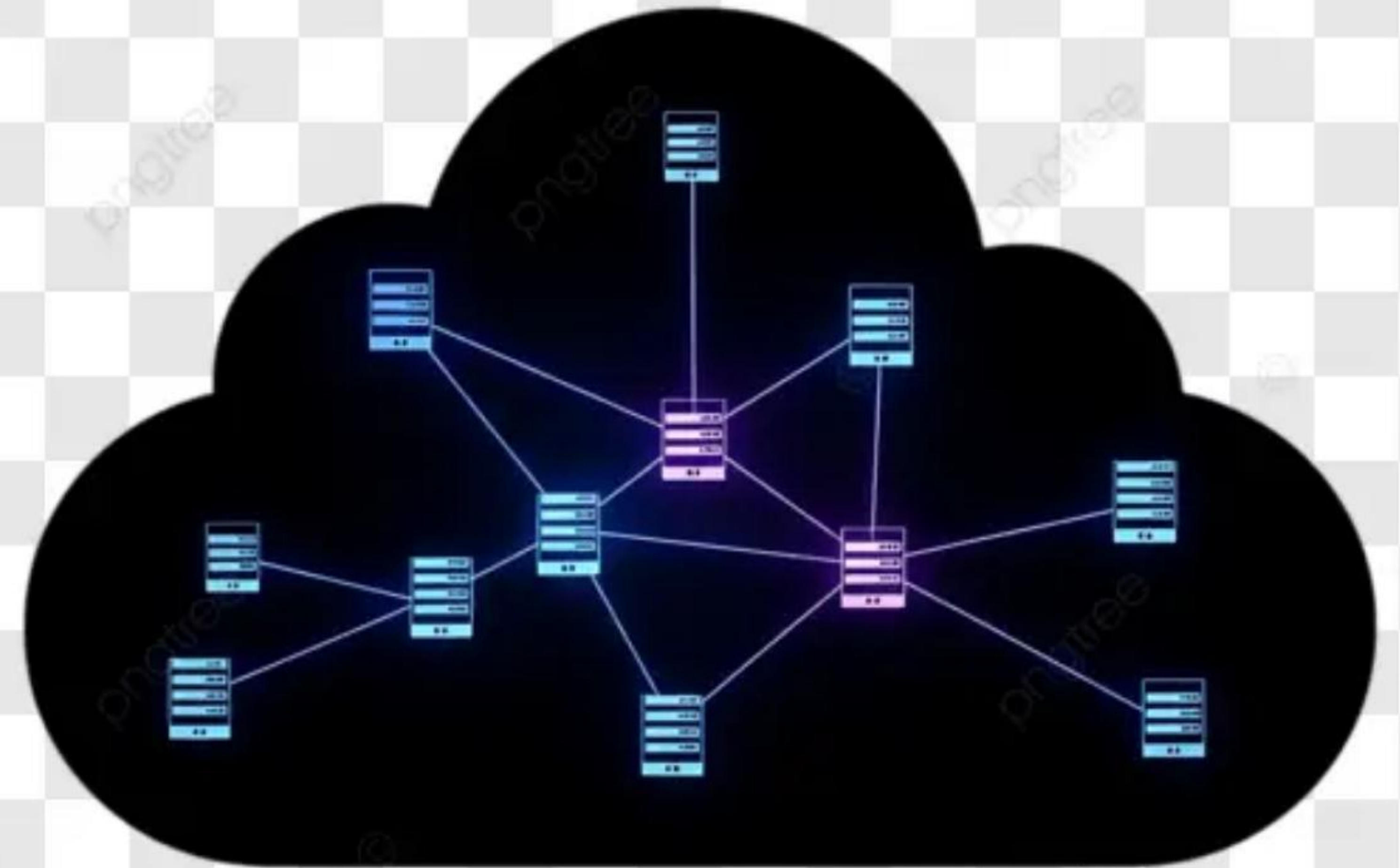


System Architecture

Decoupled & Scalable

The Eversoft integration architecture deliberately separates the high-velocity ingestion gateway from the compute-heavy AI engine.

By using a message broker and task orchestrator in the middle, the system can gracefully absorb massive traffic spikes without dropping candidate payloads, ensuring 100% data integrity.



Data Flow Diagram



Questions?

Thank you for exploring the Eversoft HRMS Architecture.

Image Sources



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